DIGITOMICSAI

Agentic AI Engineering Assignment

Autonomous Property-Discovery Agent for Home Buyers

Level: Graduate / Entry Focus: Agentic AI + Tool Use Submission: 4 July 2026

1. Context & Why This Matters

Home buyers spend an average of **three to six months** searching listings manually — cross-referencing school ratings, commute times, price history and neighbourhood safety across five or more websites. It is slow, repetitive, and exactly the kind of multi-source reasoning that an autonomous agent should own.

This assignment asks you to build a **conversational AI agent** that takes a buyer's preferences in plain English, autonomously queries the right data sources through tools, reasons over what it finds, and returns a **ranked shortlist of properties with a clear, structured justification** for each pick.

2. The Problem Statement

Build an autonomous AI agent that helps home buyers find, analyze, and shortlist properties by reasoning over live listings, neighbourhood data, and user preferences.

The agent must accept natural-language preferences, decide on its own which tools/data sources to call, combine the results, rank candidates against the buyer's stated priorities, and explain *why* each property made the shortlist.

3. What You Must Deliver

1. Working demo	A running project — hosted web app (preferred) or a runnable local app / notebook with a short screen-recording. The agent must handle a live conversation end-to end.
2. System design flow	A diagram of your architecture: user → agent → tools → data sources → reasoning → ranked output. See the reference design in Section 6.
3. Source code	A public Git repo (GitHub/GitLab) with a clear README: setup steps, environment variables, and how to run it.

4. Short write-up	1–2 pages: your approach, which tools the agent uses, how ranking works, trade offs, and what you'd improve with more time.

3. How You'll Be Evaluated

Agent autonomy & tool use	30%	The LLM genuinely decides which tools to call; not a hard coded pipeline dressed up as an agent.
Reasoning & ranking quality	25%	Sensible multi-signal ranking; justifications are accurate and tied to the buyer's stated priorities.
Working demo	20%	It runs, handles a real conversation, and doesn't fall over on follow-ups.
System design & code quality	15%	Clean tool interfaces, readable code, clear architecture matching your diagram.
Communication	10%	Clear README, honest write-up of trade-offs, live-vs-mocked data stated plainly.

We value honesty over polish. A smaller, real, well-reasoned agent beats an over-scoped demo that doesn't run. Tell us what's live, what's mocked, and what you'd do next.

4. Submission

- **Deadline:** End of day, **4 July 2026**.
- **Send:** a link to your public Git repo, a link to the live demo (or a screen-recording), and your 1–2 page write-up (PDF).
- **Reply to this email** with your submission, or to the address provided by your coordinator.
- **Questions?** Reply anytime — asking good clarifying questions is encouraged, not penalised.